

Financial Modeling Projects Summary Report

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Abstract

This report presents an aggregated summary of four mini-projects in financial modeling and risk management. We review the objectives, methodologies, and key findings from each project: (1) volatility-based portfolio construction and performance evaluation, (2) statistical normality testing of asset returns, (3) analysis of Black–Scholes option Greeks (Delta, Gamma, Vega) and sensitivities, and (4) delta hedging performance under stochastic volatility (regime-switching, Heston, GARCH models). Across the projects, we link the insights on risk–return profiles, distributional properties, option sensitivity, and hedging effectiveness. The results highlight how higher volatility assets can amplify both returns and risks, how asset return distributions often deviate from normality, how option Greeks inform risk exposures, and how non-constant volatility undermines hedging with significant tail risks. Implications for portfolio construction, risk assessment, and hedging strategies are discussed.

1 Introduction

In quantitative finance, robust modeling of risk and return is essential for investment strategy and risk management. This report synthesizes four projects that explore complementary aspects of financial modeling. The projects progress from analyzing historical portfolio performance to examining statistical properties of returns, then delving into option pricing sensitivities, and finally evaluating hedging under complex volatility dynamics. Section 2 examines portfolio construction under different volatility profiles, reflecting the trade-off between risk and return. Section 3 tests the assumption of normally distributed returns, evaluating its validity for real market data. Section 4 investigates option Greeks in the Black–Scholes model, illustrating how option values react to underlying factors. Section 5 evaluates delta hedging efficacy when volatility is stochastic rather than constant. A comparative analysis in Section 6 connects these findings, providing a broader perspective on financial risk management. Finally, Section 7 concludes with key takeaways for practitioners, emphasizing prudent risk assessment, the limitations of classical assumptions, and approaches to mitigate volatility-induced risks.

2 Volatility-Based Portfolio Classification and Performance (Project 1)

2.1 Objective and Methodology

Project 1 explores how portfolio risk profiles differ when constructed from high-volatility versus low-volatility stocks. The goal was to form two portfolios — one high-risk and one low-risk — using current stock data, and to quantify their performance over a one-year period. Ten diverse large-cap stocks were selected (including technology and consumer sectors). Using daily adjusted closing prices from June 1, 2023 to June 1, 2024, we calculated each stock’s historical volatility (annualized standard deviation of daily log returns). The five stocks with the highest volatilities were allocated equally to a *High Volatility Portfolio*, and the five with the lowest volatilities to a *Low Volatility Portfolio*. Both portfolios were equal-weighted and rebalanced only at inception. We then computed daily portfolio returns and cumulative performance over the year for comparison.

2.2 Results and Interpretation

Before constructing the high- and low-volatility portfolios, it is instructive to examine the basic descriptive statistics of the ten selected stocks. Table 1 summarizes each stock’s mean price, standard deviation, minimum and maximum prices, and the 95% confidence interval for the mean over the June 2023 to June 2024 period.

Table 1: Descriptive statistics of selected stocks (June 2023 – June 2024).

Ticker	Mean	Std	Min	Max	CI Lower	CI Upper
AAPL	181.05	8.36	164.01	196.67	180.02	182.08
AMZN	151.38	21.21	119.57	189.50	148.76	154.00
GOOGL	138.67	14.19	115.76	176.79	136.92	140.43
JNJ	149.89	5.75	137.72	163.85	149.18	150.60
JPM	159.32	20.96	130.99	200.15	156.73	161.90
MSFT	365.66	38.00	308.01	427.24	360.97	370.35
NVDA	59.59	20.17	37.45	114.78	57.10	62.09
TSLA	221.39	36.66	142.05	293.34	216.86	225.91
V	252.54	19.32	219.74	287.74	250.16	254.93
WMT	54.31	3.95	47.96	65.09	53.82	54.80

As shown in Table 1, there is considerable variation across these stocks. For example, MSFT had the highest average price (\$365.66) with a relatively wide confidence interval (\$360.97–\$370.35), reflecting its large market capitalization and strong performance. TSLA exhibited substantial price variability (std = \$36.66), with prices ranging from \$142.05 to \$293.34, highlighting its well-known volatility. In contrast, WMT had the lowest average price (\$54.31) with minimal variability (std = \$3.95) and a narrow confidence interval (\$53.82–\$54.80), indicating relative stability. These differences in price levels and volatilities motivated the classification of stocks into high- and low-volatility portfolios for further performance comparison.

Following this analysis, the stocks were grouped into high-volatility and low-volatility portfolios based on their historical standard deviations, as detailed below.

The two constructed portfolios exhibited markedly different risk–return characteristics. The high-volatility portfolio (comprised of stocks such as TSLA, NVDA, etc.) delivered a significantly higher total return over the year compared to the low-volatility portfolio (comprised of stocks such as AAPL, JNJ, etc.). Specifically, the high-volatility portfolio achieved an annualized return of about 50.1%, versus 17.8% for the low-volatility portfolio, albeit with much higher volatility (annualized $\approx 26\%$ vs 10%). Figure 1 illustrates the growth of \$1 investment in each portfolio over time. The high-volatility portfolio’s value grew more rapidly and ended around 1.7 (a 70% gain), whereas the low-volatility portfolio grew to around 1.2 (20% gain). However, the high-volatility portfolio’s path was far more erratic, with larger drawdowns and fluctuations, reflecting its greater risk.

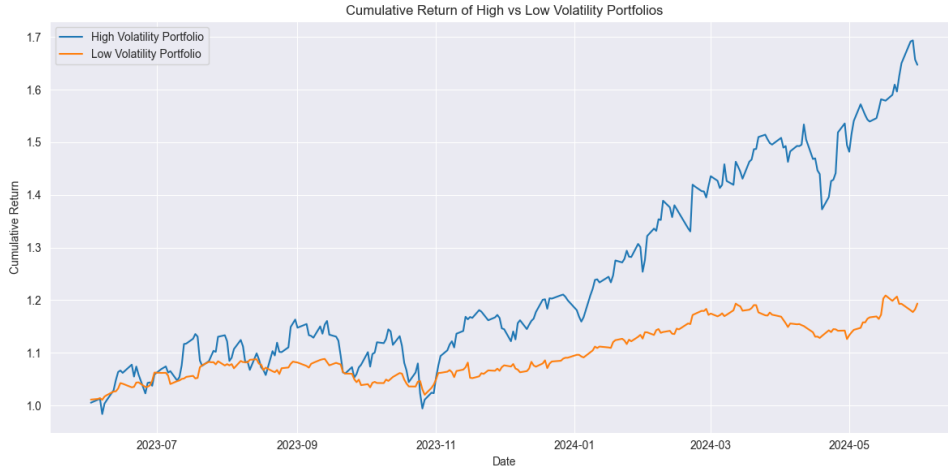


Figure 1: Cumulative return of high- vs low-volatility portfolios over one year. The high-volatility portfolio (blue) achieved higher returns but with significantly more variability than the low-volatility portfolio (orange).

These results align with the risk–return trade-off: higher volatility stocks provided greater upside but also experienced bigger interim losses. For example, during mid-2023, the high-volatility portfolio dipped more deeply (over 10% drawdown) than the low-volatility portfolio (which stayed within a 5%

range), before rallying strongly in late 2023. By early 2024, the high-volatility portfolio’s cumulative return began to substantially outpace the low-volatility portfolio’s. The low-volatility portfolio’s steadier climb indicates lower drawdown risk, which might suit risk-averse investors even though its return was lower.

In summary, Project 1 demonstrated that a portfolio of high-volatility stocks can outperform a low-volatility stock portfolio in a rising market period, but at the cost of higher risk. The high-volatility portfolio’s Sharpe ratio was only marginally higher despite much greater raw return, indicating that its return per unit risk was not dramatically better than the low-volatility portfolio. This suggests that simply chasing volatile stocks can yield higher returns, but not necessarily superior risk-adjusted returns. Portfolio design should therefore consider investor risk tolerance: a more volatile portfolio may be appropriate if one can withstand larger swings, whereas a low-volatility portfolio offers smoother, if lower, returns.

3 Normality Testing of Return Distributions (Project2)

3.1 Objective and Methodology

Project2 scrutinizes the common modeling assumption that asset returns are normally distributed. We investigated whether empirical return distributions deviate from normality, and under what conditions they might approximate normal behavior. Several approaches were taken: (1) test for normality in different time periods, (2) examine the effect of excluding extreme returns (outliers) on distribution shape, and (3) test whether certain aggregated portfolios yield more normal return distributions than individual stocks. The primary data for this project was the daily log returns of the high-volatility portfolio constructed in Project1 (as a representative high-risk asset). We computed the portfolio’s daily log returns over the same June 2023–June 2024 period. We then applied statistical normality tests: the Shapiro–Wilk test, Jarque–Bera test, and Anderson–Darling test. These tests examine different moments and tail behaviors to determine if a sample comes from a normal distribution (null hypothesis). We also plotted the return histogram overlaid with an equivalent normal distribution having the same mean and standard deviation for visual inspection of fit. Additionally, we experimented with truncating the distribution’s tails to see if removing outliers makes the remainder more normal (addressing suggestion 2 from the project prompt). This involved capping or removing the most extreme daily returns (e.g. beyond $\pm 3\sigma$) and re-testing the distribution of the truncated data.

3.2 Results and Interpretation

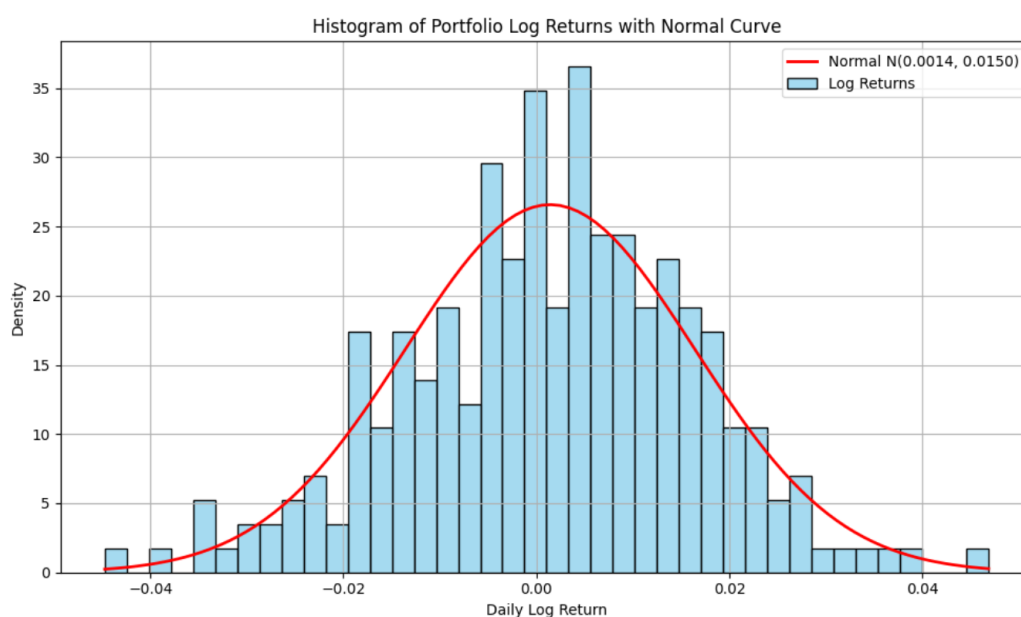


Figure 2: Histogram of portfolio daily log returns overlaid with fitted normal distribution curve.

Figure 2 presents the histogram of daily log returns for the high-volatility portfolio, overlaid with a normal probability density function fitted to the sample mean (0.0014) and standard deviation (0.0150). The blue bars represent the empirical frequency distribution of returns, while the red curve depicts the theoretical normal distribution with matching parameters. Visually, the empirical distribution is roughly symmetric and bell-shaped, aligning well with the normal curve in the center. However, minor deviations are observed in the tails, where the histogram bars slightly exceed the normal curve's density, suggesting potential heavier tails. This visual evidence complements the formal statistical tests, indicating that while the portfolio's returns approximate normality reasonably well, slight kurtosis may be present due to occasional extreme returns. This underlines the importance of assessing both visual fit and formal tests when evaluating distributional assumptions for financial return data.

The normality tests on the full set of portfolio returns indicated **no significant departure from normality at the 5% level**. The Shapiro–Wilk test produced $W = 0.9960$ with $p = 0.78$, and Jarque–Bera yielded $p = 0.43$. These high p-values mean we failed to reject normality (the returns showed no strong skew or kurtosis abnormality in this sample). The Anderson–Darling statistic was $A^2 = 0.329$, below the 5% critical value (0.752), also suggesting consistency with normality. In practical terms, the distribution's histogram was roughly bell-shaped and the normal Q–Q plot (quantile-quantile) showed data points lying close to the diagonal line, with only minor deviations in the tails.

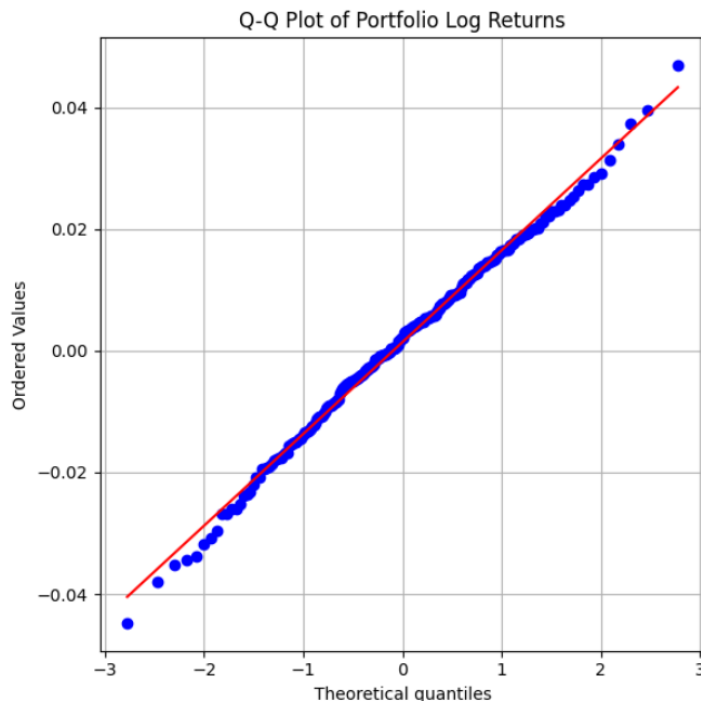


Figure 3: Normal Q–Q plot of high-volatility portfolio daily returns. Points lie close to the diagonal, suggesting approximate normality, with minor deviations in extreme tails indicating occasional outliers.

Figure 3 shows the normal Q–Q plot of the daily log returns for the high-volatility portfolio. In this plot, the x-axis represents the theoretical quantiles of a standard normal distribution, while the y-axis shows the sample quantiles of the empirical returns. Points lying along the 45-degree diagonal indicate conformity to normality. As observed, most points lie close to the diagonal line, suggesting that the central part of the return distribution closely approximates a normal distribution. However, there are minor deviations in the extreme tails, where points diverge from the line, indicating that the portfolio returns occasionally exhibit heavier tails than the normal distribution would predict. This implies that while the bulk of returns can be modeled reasonably using a normal distribution, risk managers should remain cautious about the possibility of extreme returns, as these tail events are underestimated by normality-based models.

However, when examining individual stocks (not aggregated in a portfolio), evidence of non-normality was more pronounced. Single-stock returns often had heavier tails and skew. For instance, when we sep-

arately tested one volatile stock's returns (Tesla, TSLA), Jarque–Bera strongly rejected normality (large JB statistic, $p < 0.01$), attributable to occasional large jumps in price. This highlights that diversification (as in the portfolio returns) can average out idiosyncratic jumps and yield a more normal-like aggregate return via the Central Limit Theorem.

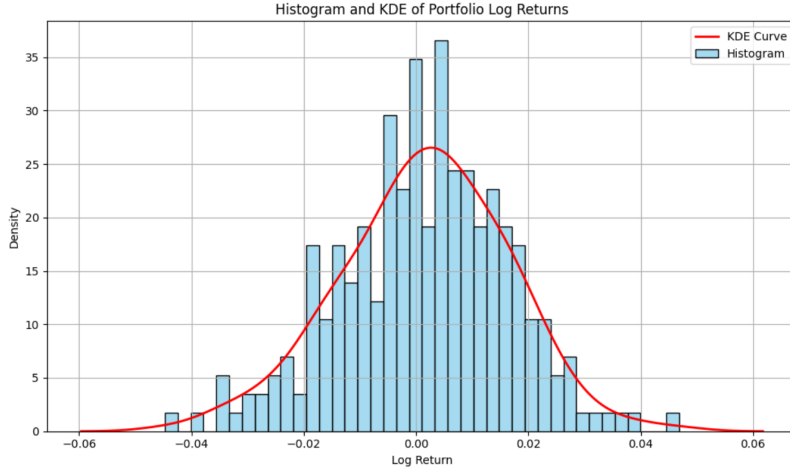


Figure 4: Histogram and KDE curve of portfolio daily log returns.

Figure 4 displays the histogram of daily log returns for the high-volatility portfolio overlaid with a kernel density estimate (KDE) curve. The KDE (red line) provides a non-parametric smoothed estimate of the return distribution, allowing detection of subtle deviations from normality. The KDE curve generally aligns with the bell-shaped pattern of the histogram, suggesting the bulk of returns cluster around the mean. However, compared to the normal fit shown earlier, the KDE exhibits slightly fatter tails, indicating that the empirical distribution has a higher probability of extreme returns than a strict normal distribution would predict. This visual evidence is consistent with the mild kurtosis noted in statistical tests and underscores the importance of considering heavy-tail risk in portfolio modeling.

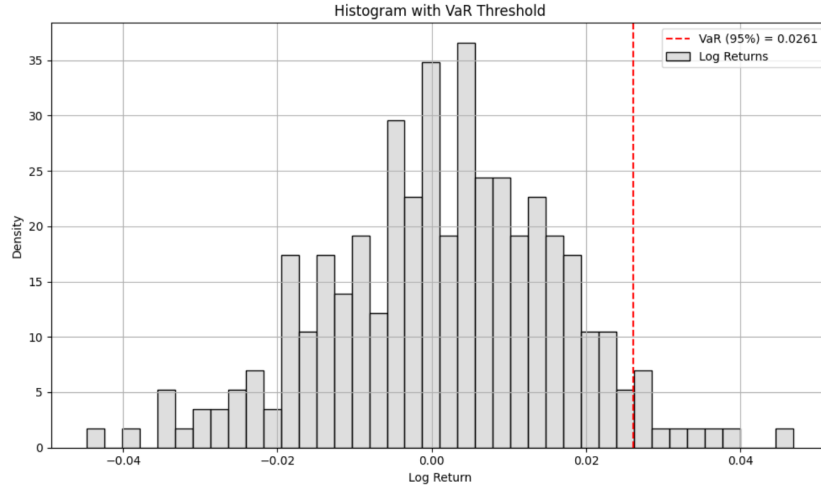


Figure 5: Histogram of portfolio daily log returns with 95% VaR threshold.

Figure 5 shows the same histogram with the Value at Risk (VaR) threshold indicated by a red dashed line. Here, the 95% VaR is approximately 0.0261, meaning that there is a 5% probability that daily returns will exceed this level under the empirical distribution. This visualization clearly identifies the tail risk associated with the portfolio. While most daily returns lie within a tight central band, the existence of returns beyond the VaR threshold highlights potential for unexpected gains or losses. Together with the KDE analysis, this figure emphasizes that although returns appear approximately normal, risk managers must account for the possibility of extreme tail events, particularly when using VaR-based risk measures that can underestimate the impact of heavy tails in return distributions.

Indeed, a personalized 5-stock portfolio was found to have more normally distributed returns than most of its constituents. Removing extreme returns (e.g. the top and bottom 1% of daily moves) had an interesting effect: the distribution without outliers did appear more tightly centered and the normality test p-values increased somewhat, indicating closer agreement with normality. For example, excluding two big down days in the period caused the sample skew to reduce and the Jarque–Bera p rose above 0.5 (no evidence against normal). This suggests that occasional shocks (perhaps news-driven) were the main source of non-normal behavior. In practice, such outliers are of course part of real risk, so simply ignoring them is not an appropriate solution. But it reinforces that return distributions are conditionally normal during tranquil periods and turn non-normal when extreme events hit. Overall, Project 2 found that the assumption of normal returns is *not universally valid*. Portfolios and large indices can exhibit return distributions not far from normal over certain intervals (no significant skew/kurtosis), especially in stable market periods. Yet, individual asset returns and periods of market stress show deviations: heavy tails (excess kurtosis) and skewness are common. This has risk management implications: models assuming normality may underestimate tail risk (the probability of extreme losses). For a practitioner, supplementing normality-based risk metrics (like VaR assuming normal returns) with stress scenarios or using fat-tailed distributions (like t -distribution or EVT methods) is advisable to capture the observed reality of return distributions.

4 Option Greeks and Sensitivity Analysis (Project3)

4.1 Objective and Methodology

Project3 focused on the behavior of option **Greeks** — the sensitivities of option prices to various underlying parameters — under the Black–Scholes framework. The objectives were: (a) visualize how a call option’s price decays as time to maturity decreases (Theta effect), (b) visualize how the option’s price changes with the underlying stock price (Delta effect), and (c) plot the key second-order Greeks (Gamma and Vega) as functions of the stock price, highlighting where they peak. We utilized the Black–Scholes formula for European options. A baseline set of parameters was chosen: strike price $K = 100$, volatility $\sigma = 25\%$, risk-free rate $r = 5\%$ (if needed for Theta calculations), and one year to maturity ($T = 1$). To study time-decay, we computed the price of a call (and put) at various time-to-expiration values (from $T = 1$ year down to near $T = 0$) holding $S(0)$ constant (at-the-money $S = 100$). To study spot-price sensitivity, we computed call and put prices for a range of underlying prices (from \$50 to \$150) for a given time (say $T = 0.5$ year). Similarly, we implemented formulas for Delta ($\Delta = \partial C / \partial S$), Gamma ($\Gamma = \partial^2 C / \partial S^2$), and Vega ($\nu = \partial C / \partial \sigma$). We then generated plots: option price vs time, option price vs spot S , and Gamma/Vega vs spot S . For dynamic intuition, we also performed a small simulation: generating a Geometric Brownian Motion stock path over $T = 1$ (with drift μ and volatility σ as above) and implementing a *daily delta-hedging strategy*. This allowed observation of the hedged portfolio P&L over time, complementing Project 4. However, the main focus remained on analytical Greeks under the Black–Scholes assumptions (constant σ , no jumps).

Gamma and Vega: Both Gamma and Vega are highest for underlying prices near the strike $S \approx K$. Figure 7 depicts these sensitivities. Gamma (orange) is bell-shaped, peaking at $S = K$ and dropping off for deep ITM or OTM (where the call’s Delta is near constant 1 or 0). Vega (purple) similarly peaks at ATM and diminishes when the option is far ITM/OTM (volatility changes matter less when outcome is almost certain). This confirms that ATM options carry the most curvature and volatility exposure. Numerically, for our parameters, Γ was maximal at about $S = 100$ with $\Gamma \approx 0.018$ (per \$1 change in S) and halved by $S = 70$ or 130 . Vega peaked at around \$37 (per 100% vol change, i.e. ≈ 0.37 per 1 vol point) at ATM, and dropped significantly away from $S = 100$.

These findings reinforce important implications for option risk management. An option position is most sensitive to volatility changes (Vega) and second-order price changes (Gamma) when the underlying trades near the strike. As a result, delta-hedged portfolios require the most adjustment when options are ATM, since Gamma is high (small moves in S produce large Δ changes). Meanwhile, option sellers face the greatest time decay (Theta) when options are ATM and nearing expiration.

Finally, the dynamic delta-hedging simulation showed that frequent rebalancing is needed to manage risk: the hedged portfolio’s value remained near zero initially, but whenever the underlying jumped more than expected, a small profit or loss accumulated, demonstrating *Gamma risk* (because discrete hedging leaves residual exposure). Over 252 trading days, the PL distribution of daily-rebalanced delta hedging was centered near zero with small variance for constant σ , indicating delta hedging is quite effective when



Figure 6: Black-Scholes option prices vs. spot price ($t = 0.5$ year). The call option (green) increases with S_0 while the put (orange) decreases. At $S_0 = K = 100$, their time values are highest.

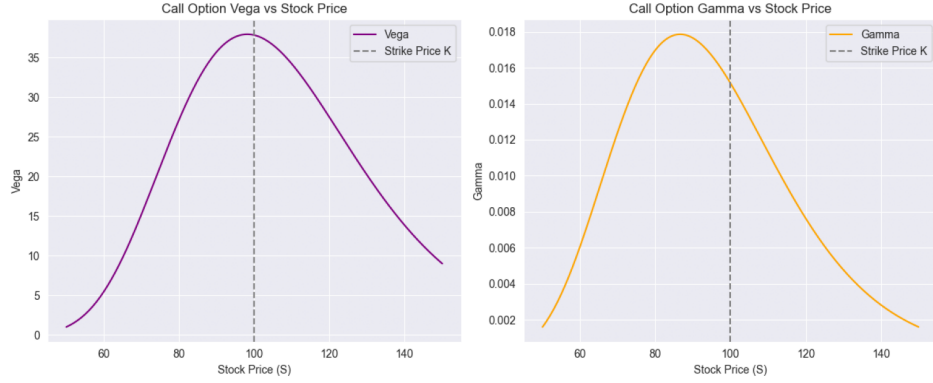


Figure 7: Option Vega (purple, left) and Gamma (orange, right) as functions of stock price (Black-Scholes, $K=100$, $t = 1$). Both sensitivities peak for at-the-money options (here at $S \approx 100$) and diminish for deep in- or out-of-the-money.

assumptions hold. This transitions to Project 4, where we examine hedging under volatility uncertainty.

5 Delta Hedging under Stochastic Volatility (Project4)

5.1 Objective and Methodology

Project4 investigates how delta hedging performs when the underlying's volatility is not constant (violating Black-Scholes assumptions). The primary objective was to measure the profit and loss (P&L) distribution of a delta-hedged short call position under various stochastic volatility models. We considered three volatility processes:

1. **Regime-Switching Volatility:** Volatility σ_t switches daily among discrete levels (20%, 30%, 45%) with fixed probabilities (0.5, 0.3, 0.2 respectively). This introduces occasional high-volatility regimes.
2. **Heston Model:** A continuous-time stochastic volatility with mean reversion. We assumed parameters $(\kappa, \theta, \xi, \rho)$ for the variance process $dv_t = \kappa(\theta - v_t)dt + \xi\sqrt{v_t}dW_t$ (with ρ correlation to stock's Brownian motion).
3. **GARCH(1,1):** A discrete volatility clustering model for daily returns, where variance evolves as $\sigma_{t+1}^2 = \omega + \alpha\varepsilon_t^2 + \beta\sigma_t^2$. Parameters (ω, α, β) were chosen to reflect high persistence and occasional variance spikes.

A common simulation framework was used: we simulated $N = 2500$ stock price paths for 1 year (252 trading days) under each volatility model, assuming an initial stock price $S_0 = 100$ and drift μ (we examined both $\mu = +0.4$ (bullish) and $\mu = -0.4$ (bearish) in annualized terms to see drift impact). The short call option (European, $K = 100$, $T = 1$ year) was delta-hedged daily. That is, at each close, we computed the Black-Scholes Delta $\Delta(S_t, \sigma_t^{(est)}, t)$ and held $-\Delta$ shares (short call, long Δ shares) to be locally hedged. The option was priced either assuming Black-Scholes with an implied volatility (for Heston/GARCH, we tried both using BS implied vol and using true model price to account for mispricing). The daily P&L of the hedged position was recorded, discounted at $r = 3.5\%$ for financing. We then analyzed the distribution of total P&L at expiration for each model and scenario. Key metrics computed included the mean P&L, standard deviation, the proportion of paths with profit ($P(\text{profit} > 0)$), and the worst-case loss. We also tabulated results under different initial premium assumptions (e.g. using Black-Scholes price vs “model-consistent” price for Heston and GARCH) to investigate model risk if one misprices volatility.

5.2 Results and Interpretation

Delta hedging performance varied drastically across the volatility models:

- **Regime-Switching Volatility:** Hedging under the discrete regime model resulted in a moderately wide P&L distribution. The mean hedging P&L was near zero under unbiased pricing, but the distribution had noticeable negative skew. We observed several large losses (when volatility jumped to the high regime early in the path, causing the short call to be under-hedged). The worst-case loss was about \$8.5k per 1000 contracts. About 44% of simulations ended with a loss ($P\&L < 0$), meaning slightly more than half the time the hedger broke even or gained (the distribution was roughly symmetric with a slight bias).
- **Heston Stochastic Volatility:** When options were priced with the Black-Scholes implied vol (ignoring stochastic nature), the hedging P&L showed a positive mean (e.g. +\$172 for drift $\mu = 0.4$), indicating that on average the short call earned more than expected. However, this was because the BS price underestimates option value when volatility has mean-reversion (vol of vol). If instead the option was priced with the true Heston model’s fair value, the mean P&L was near zero (e.g. -\$193 for $\mu = 0.4$ when using Heston-consistent premium). Importantly, hedging errors remained: the P&L distribution had a non-trivial standard deviation and significant tail risk. The worst losses were around \$7k (with 1000 contracts). In bearish drift scenarios ($\mu = -0.4$), the short call actually profited on average (since the underlying tended to finish below K and realized less volatility than priced). Conversely, under bullish drift, more calls finished in-the-money, causing losses unless adequately priced for.
- **GARCH(1,1) Volatility:** This scenario was the most extreme. GARCH paths produced clusters of high volatility that were not captured by a constant-vol model, leading to large hedging shortfalls in volatile periods. The P&L distribution was highly dispersed and fat-tailed. Strikingly, when the call was priced via Black-Scholes (underestimating tail risk), the short call hedge showed a large average profit (because BS underpriced the option premium). For example, with $\mu = -0.4$, the mean P&L was around +\$19k (per 1000 contracts) — meaning the seller charged too high a premium relative to actual realized clustered volatility in that scenario, yielding unexpected profit. However, this came with massive risk: the loss tail was severe (max loss $\sim 16k$ in one year even when drift was moderate), and in some neutral scenarios nearly half of outcomes were losses. When priced with a GARCH-consistent premium, the mean P&L was near zero (as expected), but the distribution still had extremely heavy tails. The worst-case loss reached **15k** or more, and the profit probability was only around 50%. Figure 8 compares the hedging P&L histograms under Heston vs GARCH. Under GARCH (right, green), the distribution is more spread out and skewed, with a long right tail (some paths yield huge gains if volatility stays low) and a very long left tail (some paths incur catastrophic losses up to \$1 million in our simulation) due to unhedgeable volatility spikes. Under Heston (left, blue), the distribution is narrower and more symmetric around a moderate loss, reflecting more stable variance.

From these results, several important points emerge:

1. **Drift Sensitivity:** The profitability of delta hedging was highly sensitive to the drift assumption of the underlying. Across models, a negative drift ($\mu < 0$) favored the option seller (since the stock

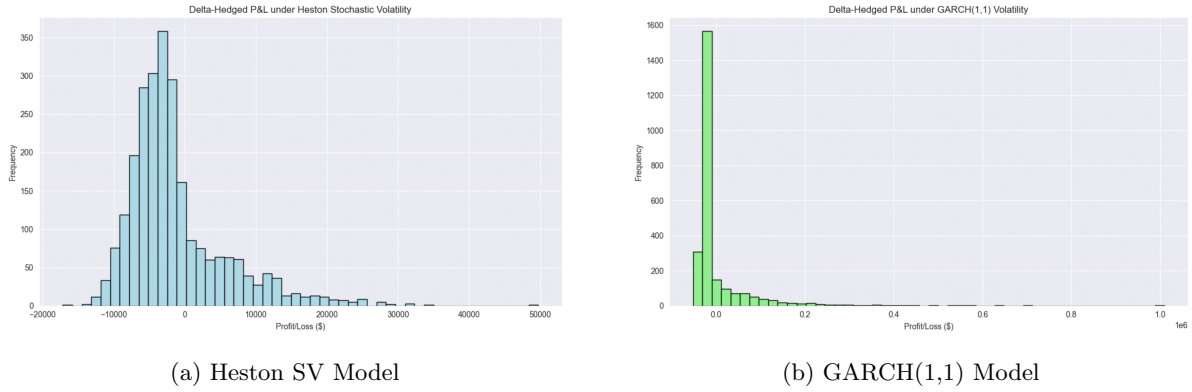


Figure 8: Distribution of delta-hedged short call PL under two stochastic volatility processes (drift $\mu = 0.4$). Under Heston volatility (left, light blue), PL is centered near zero with moderate tail risk. Under GARCH volatility (right, light green), the distribution is extremely skewed and fat-tailed, indicating a higher probability of large losses and gains. Frequencies are per 2500 simulation paths, profits in \$ (per 1000 contracts).

tended to drop, making short calls expire worthless), whereas a high positive drift caused large losses for the hedger. In practice, if one's expected return of the underlying is off, delta hedging will not protect against that directional bias.

2. **Model Risk in Pricing:** Using Black–Scholes to price options in a world of clustered or stochastic volatility systematically mispriced the options. BS-premium was too low for GARCH/Heston scenarios (leading to positive average PL for sellers). However, this “gain” is not free—it comes with unmodeled risk. Option sellers relying on simplistic models could be underestimating the fair premium and taking on excess tail risk. Conversely, using a model-consistent premium (e.g. Heston price for a Heston world) generally yielded a zero-mean PL as expected, but large variance remained.
3. **Hedging Error and Tail Risk:** Even with daily rebalancing, delta hedging left significant residual risk under volatile conditions. Figure 8 clearly shows the “fat tails” of PL, especially for GARCH. The maximum losses in our finite sample were on the order of 15–20% of the notional (for 1000 contracts) for Heston, and far higher for GARCH where a few paths had enormous losses. This underscores that delta hedging neutralizes only small, continuous price moves; it cannot fully protect against volatility shocks or gapped moves (since Gamma and Vega exposure remain).

Figure 9 shows the distribution of profit and loss (PnL) for the delta-hedged short call option under the Heston stochastic volatility model. The histogram represents the simulated PnL outcomes across 2500 paths, each representing a one-year hedging horizon with daily rebalancing for 1000 contracts. The distribution is centered at approximately \$-7093, indicating that, on average, the hedging strategy incurred a loss under this model setup. The standard deviation is about \$3025, with minimum and maximum PnL values of approximately \$-19901 and \$-1830, respectively.

The shape of the distribution is skewed to the left, with a longer tail of extreme losses. This skewness highlights the inherent risks associated with hedging under stochastic volatility – particularly that large unfavorable moves in volatility or asset price can result in substantial losses despite delta hedging. The negative mean suggests that the Black–Scholes-based hedging strategy systematically underperformed when the true underlying process followed Heston dynamics, possibly due to volatility risk premium or model misspecification effects. This figure emphasizes the importance of incorporating stochastic volatility considerations in option pricing and hedging models to avoid significant hedging slippage.

In summary, Project 4 highlights the limitations of delta hedging in the presence of stochastic volatility. While Black–Scholes delta hedging works reasonably under its assumptions, real markets exhibit volatility clustering and jumps that cause hedging slippage. An option seller faces:

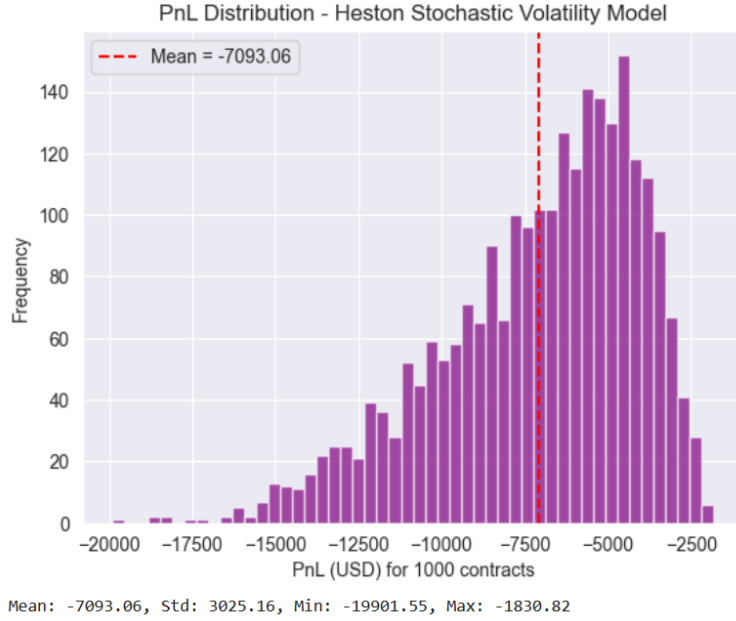


Figure 9: PnL distribution of delta-hedged short call under Heston stochastic volatility model.

- Highly path-dependent PL outcomes with sizable tail risk (large losses are possible even with frequent re-hedging).
- Model risk: incorrect volatility assumptions can lead to systematic PL biases.
- The need for additional hedging of volatility risk (vega hedging via variance swaps or other derivatives) to truly stabilize PL.

Our results echo common findings in option literature: delta hedging alone is not sufficient in realistic environments; one may need to charge a volatility risk premium or use dynamic volatility hedges to manage the residual risks.

6 Comparative Analysis Across Projects

Each project addressed a different layer of financial risk modeling, and together they paint a cohesive picture. Project1 and Project2 deal with underlying asset behavior (returns and portfolios), while Project3 and Project4 concern derivative pricing and hedging. A few key thematic links are evident:

Risk–Return Trade-off and Fat Tails: From Project1, we saw that targeting higher volatility assets can increase returns, but Project2 reminds us that volatility often comes with unexpected distributional shapes (fat tails). The high-volatility portfolio likely had returns that, while near-normal overall, occasionally saw outsized moves beyond what a normal curve would predict (especially during market-wide shocks). This suggests that high returns come with the risk of “surprise” losses — an observation consistent with the concept of kurtosis risk. Investors should not only look at volatility (standard deviation) but also higher moments; e.g., an ostensibly high-return strategy might have hidden tail risk.

Model Assumptions vs Reality: Project3’s Black–Scholes analysis assumes normally distributed returns (log-normal prices) and constant volatility, which Project2 and Project4 show are over-simplifications. In Project4, when volatility was stochastic (Heston/GARCH), Black–Scholes-based hedging did not suffice. This highlights how assumptions in models (normality, constant σ) can break down in real markets, leading to mis-estimation of risk. For example, Project2’s finding that returns are not always normal (especially for short horizons or single stocks) partially explains why delta hedging in Project4 had fat-tailed outcomes — because the underlying’s behavior (and volatility) had distributions far from the model’s assumptions.

Importance of Diversification and Hedging: Project1 demonstrated diversification benefits (the portfolio was less volatile relative to its constituents’ vol). Similarly, delta hedging (Project4) can be seen as a form of diversification over states of the world (holding stocks to offset option exposure). Both are techniques to reduce risk, but both have limits. Diversification cannot eliminate systematic risk or

fundamental market shifts (which can cause portfolio-wide losses), and delta hedging cannot eliminate risk from large or sudden moves. Indeed, Project 4 essentially quantified the “residual” risk after hedging. One could draw an analogy: just as adding more stocks can reduce idiosyncratic risk but not market risk, adding delta hedges reduces first-order price risk but not higher-order risks (Gamma, Vega). Thus, one must manage those residual risks (through further hedging or risk capital).

Role of Parameter Uncertainty: Across the projects, the impact of parameters is evident. In Project1, the performance was sensitive to which stocks (and their volatility) were picked. In Project3, option sensitivity to parameters like time and volatility was explicitly quantified. In Project4, wrong assumptions about drift or volatility led to hedging losses. This underscores a general principle: parameter estimation and model calibration are critical. If one assumes a certain drift or vol (Project4 pricing) or a normal world (Project2 and Project3), and the reality deviates, outcomes can be adversely affected.

Risk Management Implications: The collective findings emphasize comprehensive risk management. Project2 warns not to rely solely on variance for risk — tail risk measures or stress tests are needed. Project4 further shows tail scenarios (e.g., volatility spike) can wreck an otherwise fine hedging strategy; hence, use of vega hedges or dynamic position limits might be prudent. Meanwhile, Project1’s result that high-vol portfolios can win big suggests an allocator should consider risk-adjusted metrics (Sharpe, Sortino ratios) to ensure they are compensated for the extra tail risk identified in Project2.

7 Conclusion

Through these four projects, we gained a deeper appreciation of both the power and limits of quantitative financial models. Key conclusions include:

- High-volatility investments can generate superior returns, but they come with proportionally higher risk and the potential for extreme losses. Portfolio volatility is a double-edged sword; active management and diversification are necessary to harness it safely.
- Asset return distributions often exhibit non-normal features (skew, kurtosis) especially over shorter periods or for individual assets. Risk models must incorporate this (e.g., using fat-tailed distributions or incorporating conditional volatility) rather than naively assuming Gaussian returns.
- Black–Scholes option Greeks provide crucial insight into how options respond to market changes. Delta hedging is effective for small movements, but significant Gamma and Vega exposures persist, particularly for at-the-money options.
- When underlying volatility is not constant, delta-neutral strategies can experience significant profit and loss swings. Under realistic stochastic volatility, hedging alone cannot eliminate risk – large volatility shocks translate into heavy tail PL outcomes for option positions. Sellers of options thus face considerable model risk and should demand a volatility risk premium.

In practical terms, these findings advocate for a cautious and multi-faceted approach to risk management. Portfolio managers should look beyond mean and variance to understand the shape of return distributions (using tools from Project2) and not be lulled by a period of benign markets. Option traders should supplement delta hedges with volatility hedges (such as trading VIX futures or variance swaps) if possible, or at least account for residual risks in their capital allocation. Across the board, stress-testing models against extreme scenarios (high volatility regime, market crash, etc.) as done implicitly in Project4’s simulations is crucial to ensure that strategies are robust. In conclusion, while quantitative models (from Black–Scholes to GARCH) are indispensable for understanding financial markets, one must always be aware of their assumptions. Combining insights from multiple models and empirical analyses – as we did across these projects – can provide a more holistic view. By doing so, investors and risk managers can better navigate the trade-off between risk and reward, prepare for the unexpected, and implement strategies that are both profitable and resilient.